DATA ANALYSIS **01/10/25**

1. The table below shows the amounts spent by Premier League clubs during the Summer 2020 transfer window as per football365.com. Some transfers have been excluded.

Amount *x* million pounds	Frequency		
$0 < x < 5$	16		
$5 \leq x < 10$	8		
$10 \leq x < 15$	7		
$15 \leq x < 20$	16		
$20 \leq x < 30$	14		
$30 \leq x < 40$	5		
$40 \leq x < 50$	5		
$50 \leq x < 75$	2		

   a. What sort of skew would you expect this to exhibit if it were plotted as a histogram?
   b. Why is a histogram more appropriate than a bar chart to display these data?
   c. Calculate estimates for the mean and median and comment on your answer to a.
   d. Suggest why some transfers have been excluded.

2. The stem and leaf chart aside shows the number of goals scored by each Premier League team in the 2019-20 season:

   a. What sort of skew do these data show?
   b. Why would this be the skew we might expect to see?
   c. Find the mean number of goals scored by a Premier League team in this season.
   d. Work out the five-point summary for these data.

20	6
30	1 6 8 9 9
40	0 1 3 4 9
50	1 1 6
60	1 6 7 9
70	
80	5
90	
100	2

Key: 10|1 = 11

The diagram below shows the above data split by goals home and away:

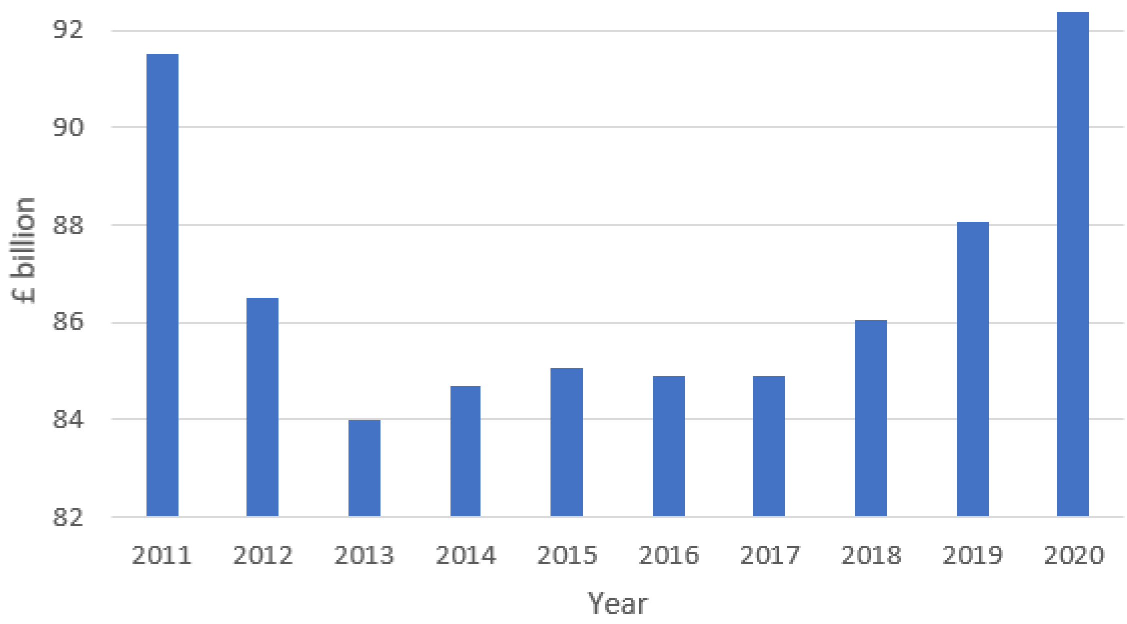
Home		Away
	0	7
9 5	10	4 5 6 8 8 9 9 9 9
7 4 4 4 2 2 2 1 0 0	20	0 0 4 5 6
6 6 5 0 0	30	0 2 3 9
0	40	5
7 2	50	

Key: 30|9 = 39

   e. Discuss the claim that teams always do better at home than away. You may find the following useful:

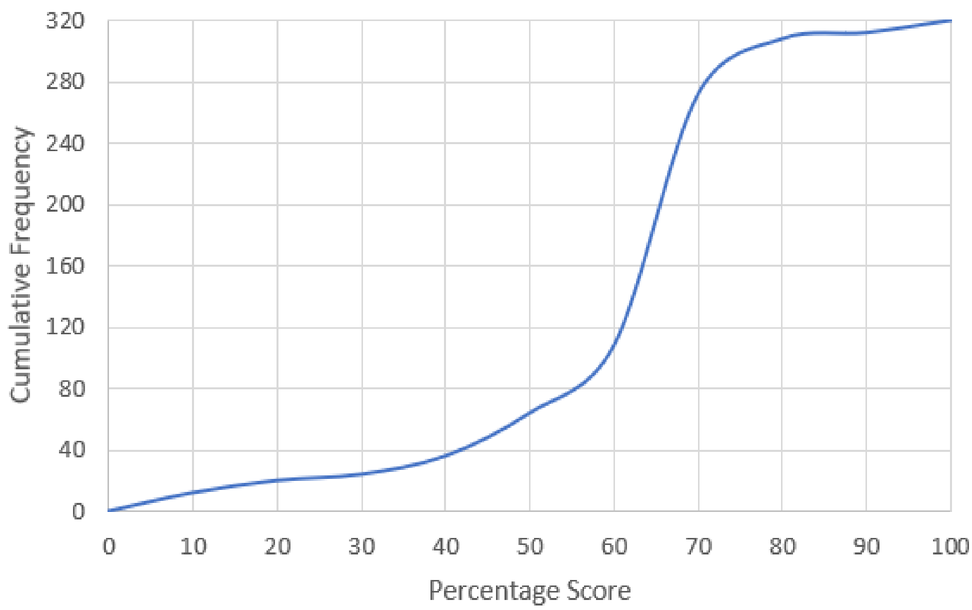
	Mean	Median	IQR
Home	28.8	24	14
Away	22.9	19.5	10

3. The bar chart below shows Government spending on Education in the years since the 2010 General Election.



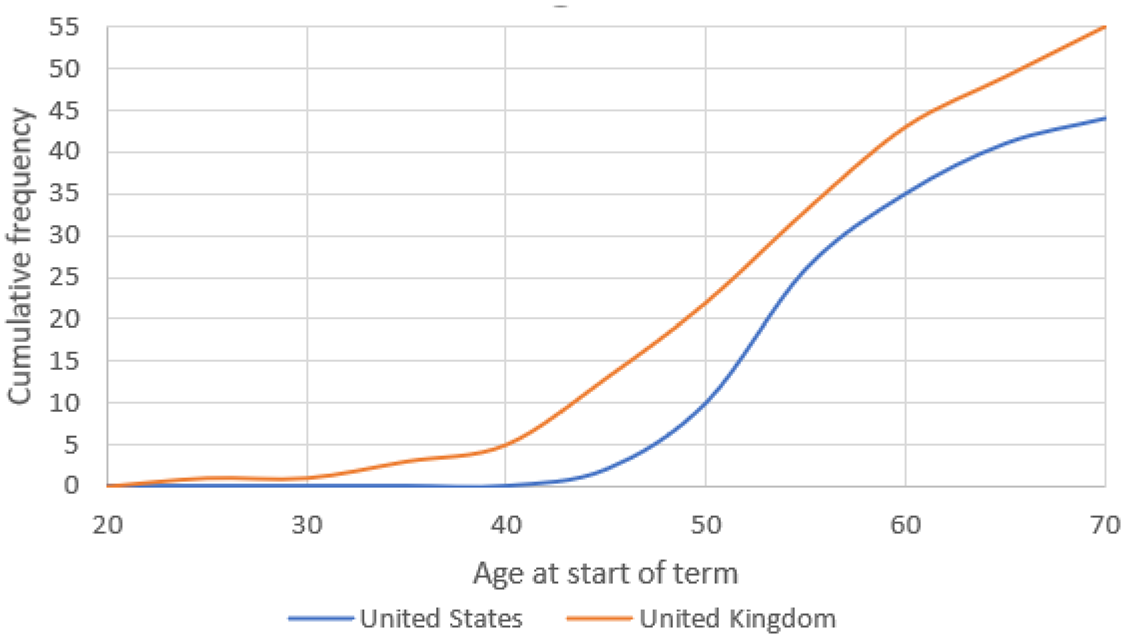
   a. Why is this bar chart misleading?
   b. Why would a histogram probably not be appropriate here?
   c. Who do you think produced this bar chart and why?

4. The cumulative frequency curve below shows the scores attained by a year group in a large school in a mathematics exam.



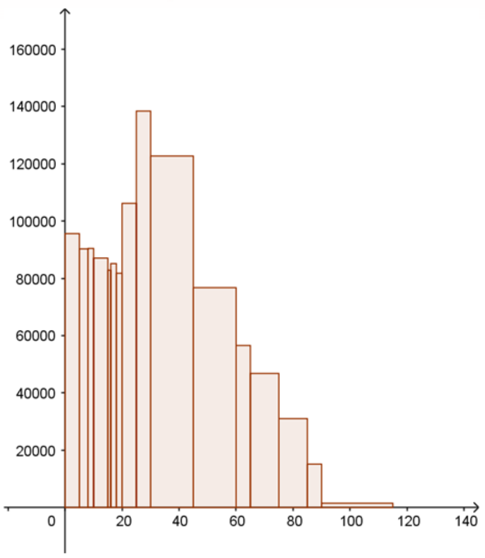
   a. Find estimates for the median and the upper and lower quartiles.
   b. Estimate how many students scored between 50% and 65%.
   c. The school decides to set the grade boundaries such that 85% of the students pass. Estimate the pass mark.
   d. What type of skew does this data exhibit?
   e. Explain why your answers to a, b and c are estimates.

5. The diagram below shows the ages of the Presidents of the United States and Prime Ministers of the United Kingdom at the time they assumed office:

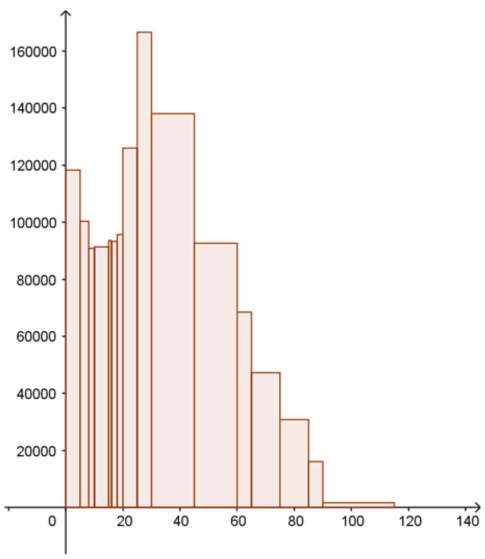


   a. It has been suggested that, on average, British Prime Ministers tend to be older than their American counterparts. Does this data support that claim?
   b. Do either of the data sets seem to be skewed? Does this change your answer to the above?

6. The diagrams below show the distributions of ages of people in London as measured by the UK Census of 2001 and 2011.



Age Structure in London 2001



Age Structure in London 2011

   a. Which age category contains the largest number of people in both diagrams?
   b. Estimate how many people we aged between 45 and 60 in London in 2001.
   c. Estimate how many people were aged between 65 and 75 in 2001 and between 75 and 85 in 2011. Why are these answers not the same?
   d. Describe the changes in the makeup of the population between 2001 and 2011 and suggest reasons for these changes.

7. The table below shows the total number of people in London's 33 Local Authorities who described their main method of travel to work as using a bus/minibus in the 2011 UK census.

263	7,339	8,347	8,571	8,926	8,951	9,215	9,553	9,725	9,950	10,898
11,115	13,024	13,395	14,314	15,283	15,476	15,789	16,779	17,728	18,336	19,677
20,424	20,959	21,595	22,372	22,660	22,859	24,193	25,922	30,541	31,873	37,956

a. One of the Local Authorities in London is the City of London. Identify which one it is and give a reason for your choice.

The *midrange* of a data set is the median average of the highest and lowest value.

b. What is the midrange for the above data?

c. In what situations might the midrange be a useful measure of central tendency?

d. Find the upper and lower quartiles for this data set and hence find the inter quartile range.

e. Identify any value(s) in the above that might be candidates to be considered an *outlier(s)*.

In fact, we define a point to be an outlier if it more than one and a half interquartile ranges above the upper quartile or below the lower. (There is another definition using something called the *standard deviation* about which more later.

f. Identify whether there are any outliers in the above data set.

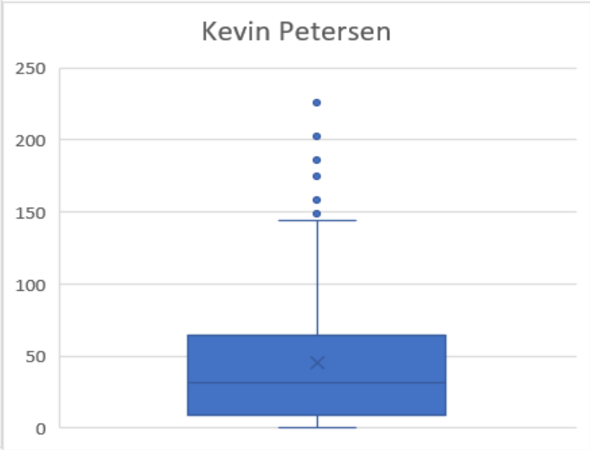
8. The table shows the numbers of people travelling to work by underground, metro, light rail or tram in the Local Authorities of Yorkshire and the Humber as per the 2011 UK census

Barnsley	115	**Harrogate**	118	**Rotherham**	450
Bradford	261	**Kingston upon Hull, City of**	51	**Ryedale**	24
Calderdale	80	**Kirklees**	151	**Scarborough**	35
Craven	25	**Leeds**	454	**Selby**	28
Doncaster	114	**North East Lincolnshire**	51	**Sheffield**	8,186
East Riding of Yorkshire	103	**North Lincolnshire**	33	**Wakefield**	116
Hambleton	37	**Richmondshire**	29	**York**	96

a. Verify that Sheffield is an outlier as per the definition above.

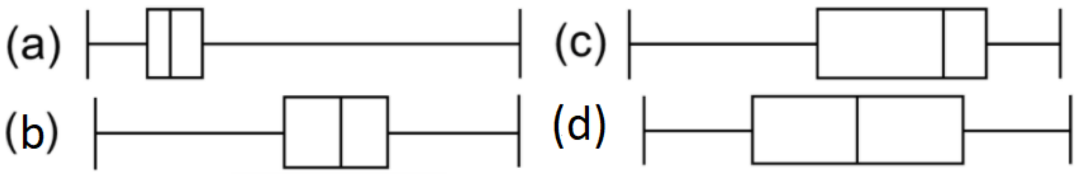
b. A member of the transport department wants to justify cutting funding for tram services in the region by saying "Apart from Sheffield, which is a clear anomaly and so should be ignored, no one really uses these service so we should stop paying for them." Evaluate his argument.

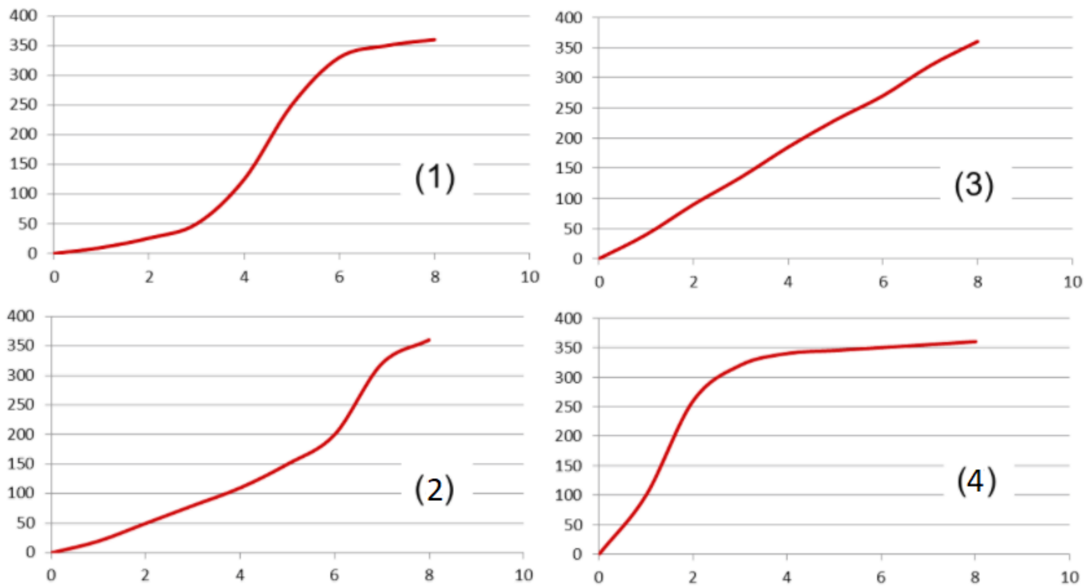
9. The diagrams below show the innings scores achieved by Alistair Cook and Kevin Petersen throughout their careers. The data is courtesy of howstat.com



a. Explain why it is not surprising that both boxplots show positive skew.
b. Explain what is shown by the dots and crosses in both diagrams.
c. “Kevin Petersen has the higher median average and so is the more accomplished batsman.” Evaluate that statement.

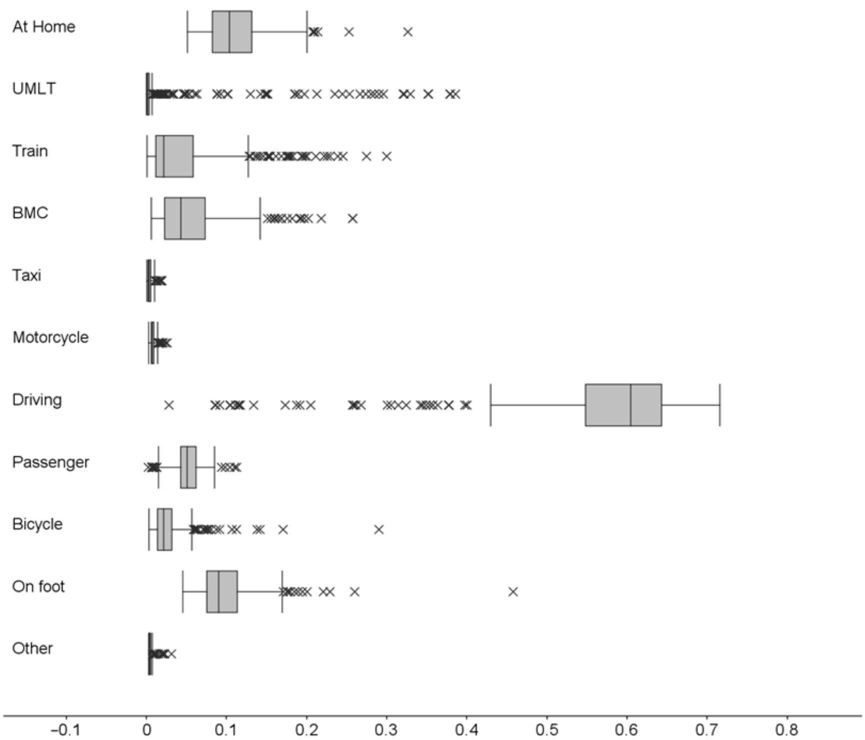
10. Match the four cumulative frequency curves below to their corresponding boxplot:



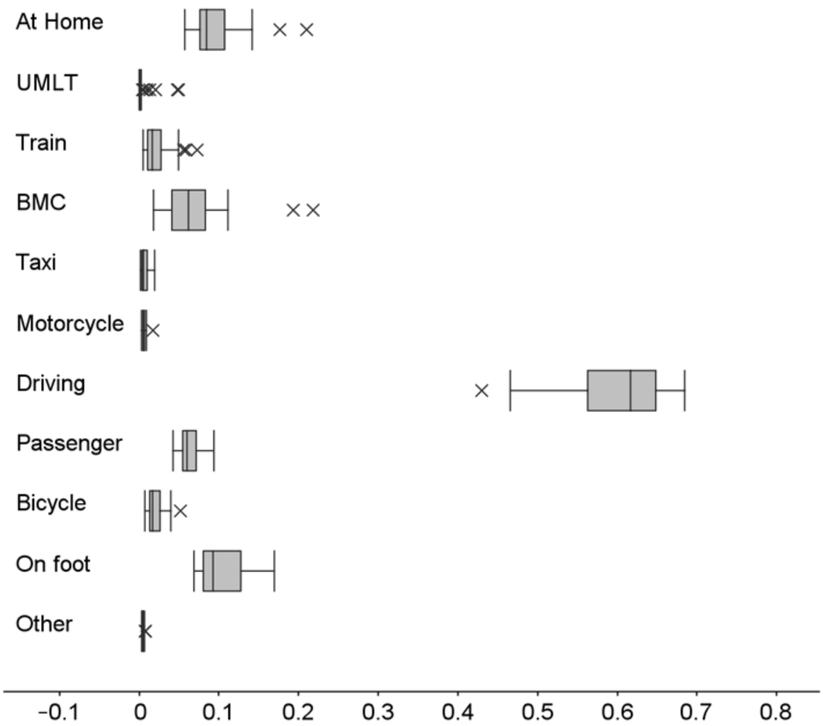


11. The stacked boxplots below show data from Method of Travel to Work in 2011 for the whole county and then for one Region.
    a. On average, what proportion of the population walks to work?
    b. Why are there fewer crosses in the North West Data?
    c. To what extent does ULMT (underground, light rail, metro and tram) use in the North West well represent the use in the population as a whole?
    d. Why do you think there are multiple crosses in the driving boxplot for the UK but not for the North West? (Do not just rehash your answer to a here, you should be more specific)
    e. Why do some of the boxplots look like vertical lines?
    f. What main change might you expect to see if the 2021 census data were available now?

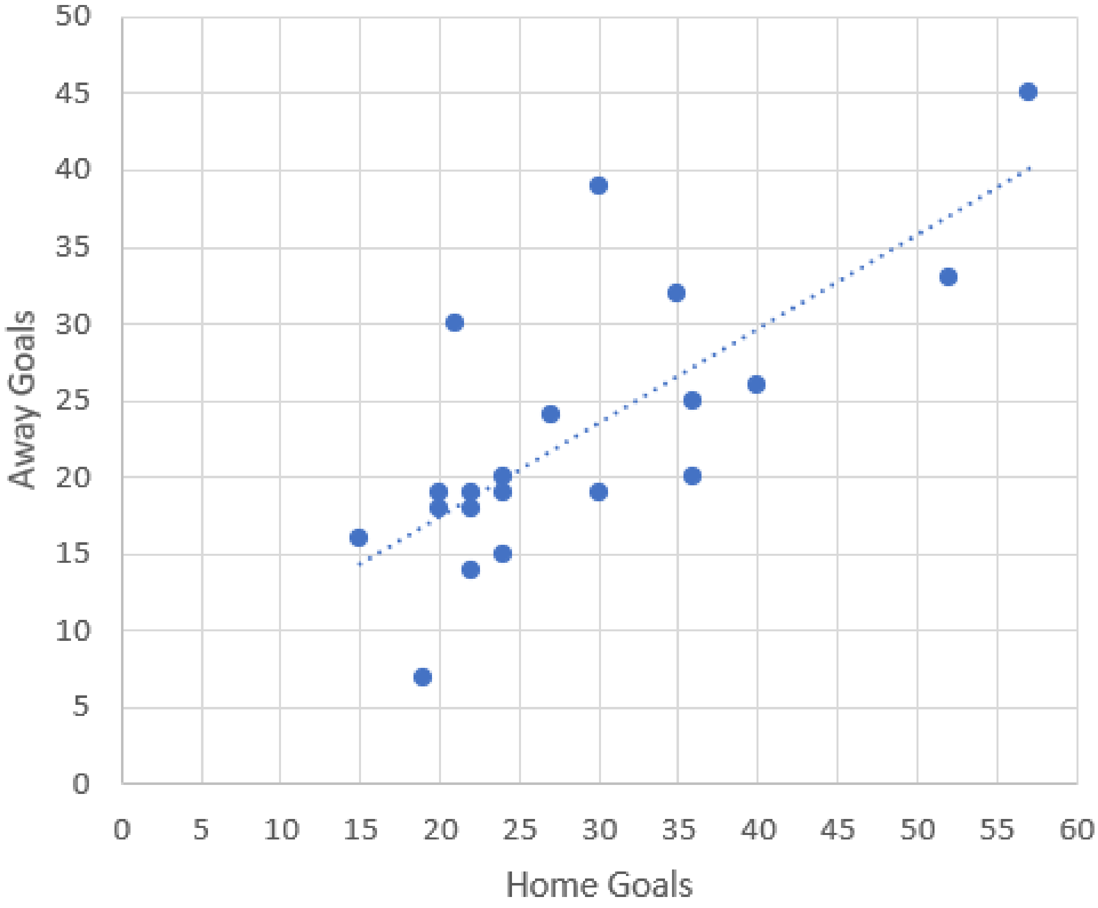
Whole population:



North West regions:



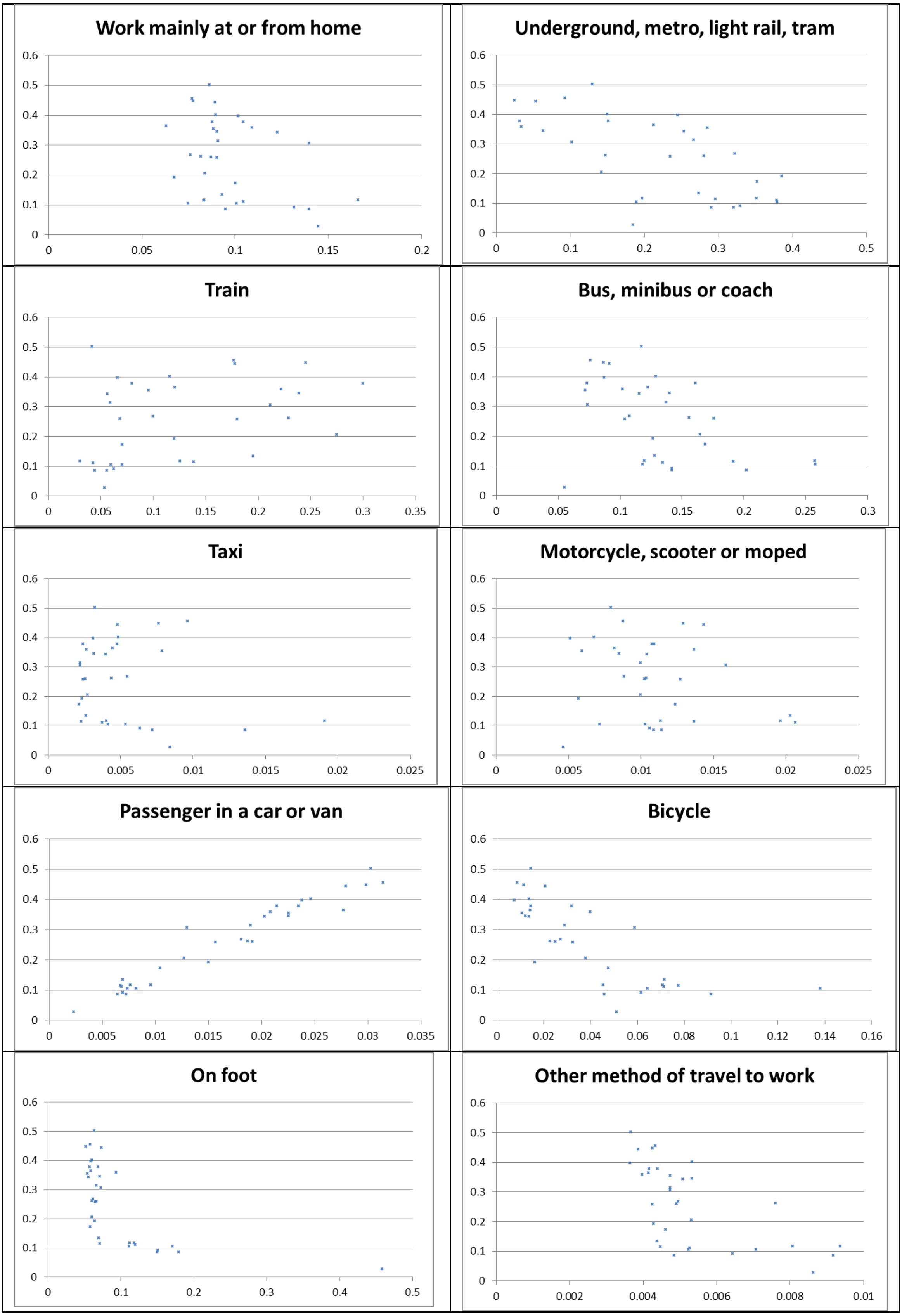
12. The diagram below shows the same data as question 2 but this time plotted as a scatter graph.



a. Describe the correlation shown by this diagram and explain why this answer is not surprising.
b. Wolverhampton Wanderers scored 27 goals at home during the 2019-20 season. How many goals did they score away from home?
c. Imagine that a team's data are missing but that you know they scored 40 away goals respectively. How many home goals might you have expected them to score?
d. Why might you not use this diagram to predict the home goals scored by a team which scored 60 goals away from home?
e. At the time of writing, Chelsea are on course to score 44 goals at home this season. Why might you not use the graph above to predict how many goals they will score away?

13. The diagrams on the next page each show the proportion of people who travel to work by driving a car or van plotted against the proportion who travel to work using each of the other 10 methods. This data is taken from Method of Travel to Work for the London Boroughs in 2011.

a. For each of the diagrams identify the type of correlation shown (if any).
b. What is the wrong with the following statement? "Greater levels of car use cause fewer people to use the underground, metro, light rail or tram."
c. Explain the negative correlation between car use and bicycle use
d. By referring to the diagrams in Question 11, explain why almost all of the relationships exhibit either no correlation or negative correlation.
e. Explain the reasons for the positively correlated relationship.
f. Identify any outliers.
g. Assume that there is another Local Authority which has a very high incidence, say 0.4, of people working from home. Why would you not use the below to estimate the incidence of car use?


Work mainly at or from home
0 0.1 0.2 0.3 0.4 0.5 0.6
0 0.05 0.1 0.15 0.2
Underground, metro, light rail, tram
0 0.1 0.2 0.3 0.4 0.5 0.6
0 0.1 0.2 0.3 0.4 0.5
Train
0 0.1 0.2 0.3 0.4 0.5 0.6
0 0.05 0.1 0.15 0.2 0.25 0.3 0.35
Bus, minibus or coach
0 0.1 0.2 0.3 0.4 0.5 0.6
0 0.05 0.1 0.15 0.2 0.25 0.3
Taxi
0 0.1 0.2 0.3 0.4 0.5 0.6
0 0.005 0.01 0.015 0.02 0.025
Motorcycle, scooter or moped
0 0.1 0.2 0.3 0.4 0.5 0.6
0 0.005 0.01 0.015 0.02 0.025
Passenger in a car or van
0 0.1 0.2 0.3 0.4 0.5 0.6
0 0.005 0.01 0.015 0.02 0.025 0.03 0.035
Bicycle
0 0.1 0.2 0.3 0.4 0.5 0.6
0 0.02 0.04 0.06 0.08 0.1 0.12 0.14 0.16
On foot
0 0.1 0.2 0.3 0.4 0.5 0.6
0 0.1 0.2 0.3 0.4 0.5
Other method of travel to work
0 0.1 0.2 0.3 0.4 0.5 0.6
0 0.002 0.004 0.006 0.008 0.01